

VERITY PORTAL

Business Specification Document: Data Integrity & Compliance Engine

1. EXECUTIVE SUMMARY

Verity Portal is a production-grade internal tool designed to solve the "Excel Engineering" crisis within complex corporate environments, such as defense and aerospace. The platform provides a centralized hub for reconciling disconnected data silos (HR, IT, Finance) and identifying high-stakes compliance violations. Built on **Hexagonal Architecture**, Verity Portal ensures that core business logic is isolated, durable, and auditor-ready.

2. BUSINESS PROBLEM & OBJECTIVES

2.1 The Problem

Current back-office operations rely on manual spreadsheet cross-referencing (VLOOKUPs/Stare-and-Compare). This process is:

- **Error-Prone:** Human error leads to ITAR, CMMC, and financial billing violations.
- **Inconsistent:** Fragmented data sources (Silos) result in conflicting records.
- **Inefficient:** Thousands of hours are lost to manual data cleaning and reconciliation.

2.2 Key Objectives

- Automate the identification of compliance gaps (ITAR, Security Clearance, Billing).
- Establish a "Single Source of Truth" by bridging disparate data silos.
- Provide non-editable, auditor-ready PDF reports to satisfy federal requirements (CMMC/ITAR).

3. PRODUCT SCOPE: THE HIGH-IMPACT FIVE

Scenario	Business Value	Compliance Target
ITAR & Export Control	Ensures restricted data is only accessed by eligible citizens.	ITAR / EAR
Clearance & Training Watchdog	Prevents work on contracts without valid security credentials.	Gov. Security Standards
Leaver/Mover Audit	Identifies unauthorized system logins post-termination.	CMMC 2.0 (AC.L2-3.1.4)
IT Asset & PO Audit	Matches procurement spend to physical inventory in use.	Financial Stewardship
Labor Billing Audit	Ensures billed labor categories match contractual mandates.	DCAA / Contract Law

4. FUNCTIONAL REQUIREMENTS

4.1 Intelligent Mapping (Input)

The system shall provide a **Shared Mapper Engine** that allows users to link "messy" legacy spreadsheet headers to standardized system attributes. It must utilize fuzzy matching logic to suggest potential pairings (e.g., "Emp_ID" to "Employee_ID").

4.2 Compliance Core (Processing)

The **Domain Layer** shall process uploads using isolated business rules. It must detect mismatches between "Source of Truth" files (e.g., HR Master) and "Activity" files (e.g., Access Logs).

4.3 Resolution & Export (Output)

- **Compliance Dashboard:** Visual indicators for high-risk violations.
- **PDF Audit Defense:** Formal reports summarizing the audit logic and results for federal examiners.
- **Sanitized CSV:** A downloadable file containing only verified rows for back-system updates.

5. TECHNICAL ARCHITECTURE & CONSTRAINTS

Architecture Model: Hexagonal (Ports & Adapters)

- **Frontend:** Angular 17+ (High-density, professional Material Design).
- **Backend:** Python 3.11+ (FastAPI for speed, Pandas for data processing).
- **Integration:** File-based ingestion (CSV/XLSX) to support air-gapped/firewalled environments.
- **Theme:** BAE Navy (#002244) primary with high-contrast functional color tokens.

6. NOT DOING (STRATEGIC OUT-OF-SCOPE)

- **Real-time API Writebacks:** The MVP will not write directly to legacy SAP/ERP systems due to security barriers.
- **Mobile Application:** The tool is optimized for desktop data analysts; mobile is not a requirement.
- **Full ERP Functionality:** Verity Portal is an *auditing* layer, not a replacement for primary databases.